FATIMA TAHIR

Software Engineer | Full-Stack Developer | AI/ML Engineer

Lahore, Pakistan fatimahtahir452@gmail.com GitHub LinkedIn Portfolio

PROFESSIONAL SUMMARY

Results-driven Software Engineering student with hands-on experience across full-stack development, AI and machine learning anchored by strong computer science fundamentals in data structures, algorithms, Object-Oriented Programming, and SOLID design principles. Experienced in architecting REST APIs, reducing query execution times by 35% using hybrid ORMs, and deploying cloud pipelines via automated CI/CD. Adaptable, detail-oriented engineer eager to apply end-to-end development skills across the full software development lifecycle.

EDUCATION

BS Software Engineering 2023 – 2027
Punjab University College of Information Technology (PUCIT)

FSc Pre-Engineering 2021 – 2023
Government Girls Higher Secondary School

EXPERIENCE

Web Development Intern June 2026 – August 2026
FentixTech

- Translated UI/UX designs into pixel-perfect, interactive web applications built completely from the ground up.
- Engineered fully responsive interfaces that adapt flawlessly across 3 viewport classes (desktop, tablet, and mobile).
- Managed full build-to-deployment lifecycles, delivering 100% of project deliverables on schedule.

TECHNICAL SKILLS

Languages: C#, C++, SQL, JavaScript (ES6+), Python, HTML5, CSS3

Backend: .NET 8, ASP.NET Core (MVC & Web API), Entity Framework Core, Dapper, LINQ, RESTful APIs, Clean Architecture, Microservices, Repository Pattern, Middleware, Dependency Injection, SignalR, JWT Authentication, ASP.NET Core Identity, Role- & Policy-Based Authorization, AJAX, CORS

Frontend: React.js, Blazor, Bootstrap 5, Tailwind CSS, jQuery, Framer Motion, Vite, Responsive Design, Figma-to-Code (UI/UX Implementation)

Databases: PostgreSQL, SQL Server, Azure SQL Database, Database Design & Normalization, Query Optimization

Cloud & DevOps: Azure App Service, Docker, Vercel, GitHub Actions (CI/CD), Git, GitHub, Postman, Swagger / OpenAPI

Machine Learning & AI: Scikit-learn, XGBoost, Neural Networks (FCNN, RNN, LSTM, GRU), NLP, LangChain, Time-Series Analysis, Feature Engineering, Model Evaluation

Testing: Unit Testing, Integration Testing, Postman API Testing

CS Fundamentals & Practices: Data Structures, Algorithms, OOP, Software Design Patterns, SDLC, Problem Solving, Debugging

WEB DEVELOPMENT PROJECTS

MediBlood-Connect – ASP.NET MVC, Azure SQL, Dapper, EF Core, SignalR, Azure App Service GitHub | Live

- Engineered and deployed a production-grade blood donation portal on Azure App Service and Azure SQL, processing patient-donor matches across 3 role-based portals (Donor, Patient, Admin).
- Implemented a hybrid data access layer combining Entity Framework Core (Identity & migrations) with Dapper micro-ORM for parameterized queries, cutting search query execution times by 35%+.
- Architected role- and policy-based access control with ASP.NET Core Identity, securing 3 core workflows: donor availability toggles, blood request submissions, and administrative approvals.
- Designed a 4-table normalized Azure SQL schema, applying Repository Pattern and Dependency Injection for strict separation of concerns and 100% parameterized SQL injection prevention.
- Integrated real-time notifications across all 3 portals via SignalR and automated GitHub-to-Azure deployment pipelines.

Scriblia – Full Stack Stationery E-Commerce Platform – React.js, ASP.NET Core Web API, EF Core, SQL Server, JWT, Vercel, Azure GitHub | Live

- Architected a full-stack e-commerce platform serving 15+ RESTful API endpoints using ASP.NET Core Web API, Entity Framework Core, and Azure SQL Server.
- Developed a responsive React frontend on Vercel with 4 key UX features: category filtering, instant search, product image zoom, and Framer Motion page transitions.
- Secured 3 protected user flows (registration, login, checkout) with JWT authorization and ASP.NET Identity role-based access.
- Implemented stateful cart and checkout modules with automated inventory deduction, keeping transactional data consistent across 4 core database tables.
- Engineered end-to-end admin CRUD covering 3 operations: real-time product listing, catalog updates, and order fulfillment status.

FURNITUR – Furniture E-Commerce Website – HTML5, CSS3, JavaScript (ES6+), Figma GitHub | Live

- Built a fully responsive e-commerce site in vanilla JavaScript from a high-fidelity Figma design: 7 pages, one shared stylesheet, and 3 breakpoints (mobile, tablet, desktop).
- Implemented dynamic product rendering with 5-dimension filtering (category, material, size, price range, sale status), sorting, pagination, and grid/list view toggle.
- Developed a reusable localStorage cart module (add, update, remove, totals) shared across 5 pages with a live item-count badge.
- Engineered a 4-step checkout flow with form validation, dynamic shipping pricing, payment toggle, and an order review page that rehydrates saved state.
- Applied event delegation and query-parameter routing to keep dynamic elements interactive in a static, backend-free project.

Libsys Pro – Full Stack Library Management System – React, ASP.NET Core Web API, Clean Architecture, Repository Pattern, Vercel GitHub | Live

- Architected a decoupled backend using 4-layer Clean Architecture (Domain, Application, Infrastructure, Web API), fully isolating business rules, HTTP controllers, and data access.
- Implemented 10+ RESTful API endpoints across 3 core modules following use-case driven design.
- Enforced business rules in the Application layer, managing book state transitions, borrow history tracking, and active status updates.
- Used in-memory data structures to isolate architecture testing, enabling rapid validation of API request flows through Swagger and Postman.

MACHINE LEARNING & AI PROJECTS

Human Activity Recognition System (KU-HAR) – Python, Scikit-learn, XGBoost, LSTM, GRU, Time-Series Analysis GitHub

- Built an activity classifier on the KU-HAR dataset (90 participants, 6-channel accelerometer and gyroscope signals, 300 readings per channel) to recognize 6 activities: standing, sitting, lying, walking upstairs, walking downstairs, and running.
- Engineered statistical and temporal features (mean, spread, min/max) and trained 4 classical baselines (Logistic Regression, SVM-RBF, Random Forest, XGBoost) on an 80/20 train-test split.
- Designed and compared 4 neural models (Fully Connected NN, Simple RNN, LSTM, GRU) with dropout and regularization, evaluated on Accuracy and F1-score.
- Found that LSTM and GRU outperformed all 6 other models by capturing temporal patterns, and analyzed the effects of class imbalance and limited data under memory constraints.

StudyPilot – AI-Driven Socratic Learning & Parent Analytics Platform (FYP – Ongoing) – React.js, .NET, Python, LangChain, PostgreSQL GitHub | Prototype

- Architecting a 3-tier AI platform (React frontend, .NET backend, Python ML inference service) for personalized Socratic learning and parent monitoring analytics.
- Developing RESTful endpoints in ASP.NET Core Web API (.NET 8) with Entity Framework Core to manage 3 domains: user profiles, task scheduling, and learning analytics.
- Integrating ML and NLP models (Python/Scikit-learn) to analyze student performance, predict focus bottlenecks, and adapt study plans dynamically.
- Building a React.js and Tailwind CSS dashboard tracking 3 metrics: learning progress, deadline urgency, and daily study streaks.
- Implementing JWT-based authentication across all 3 tiers and secure data pipelines between them.